# Body Bias Control for a Coarse Grained Reconfigurable Accelerator Implemented with Silicon on Thin BOX Technology

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# Outline

- Background
- SOTB Technology
  - SOTB CMOSFET
  - Characteristic of SOTB
- CMA Architecture
  - Concept of CMA Architecture
  - Block-diagram of CMA
- CMA-SOTB
- Real chip evaluation
  - Leakage power
  - Maximum operation frequency
  - Energy efficiency
- Policy of Body bias control
- Conclusion

# Background

- The highly energy efficient operations are required for sophisticated mobile devices.

  → Energy efficient accelerators are essential.

- A coarse-grained reconfigurable processing array(CGRA) is a promising candidate for such accelerators

  → Using highly parallel processing on a large processing element (PE) array

- However, such architectures have a weak point: large leakage power for a large PE array.

# What is SOTB technology?

- A silicon on insulator (SOI) technology

  → The leakage current and delay of the transistors can be controlled by using the body bias.

LEAP developed a novel SOI CMOS technology called silicon on thin BOX (SOTB)

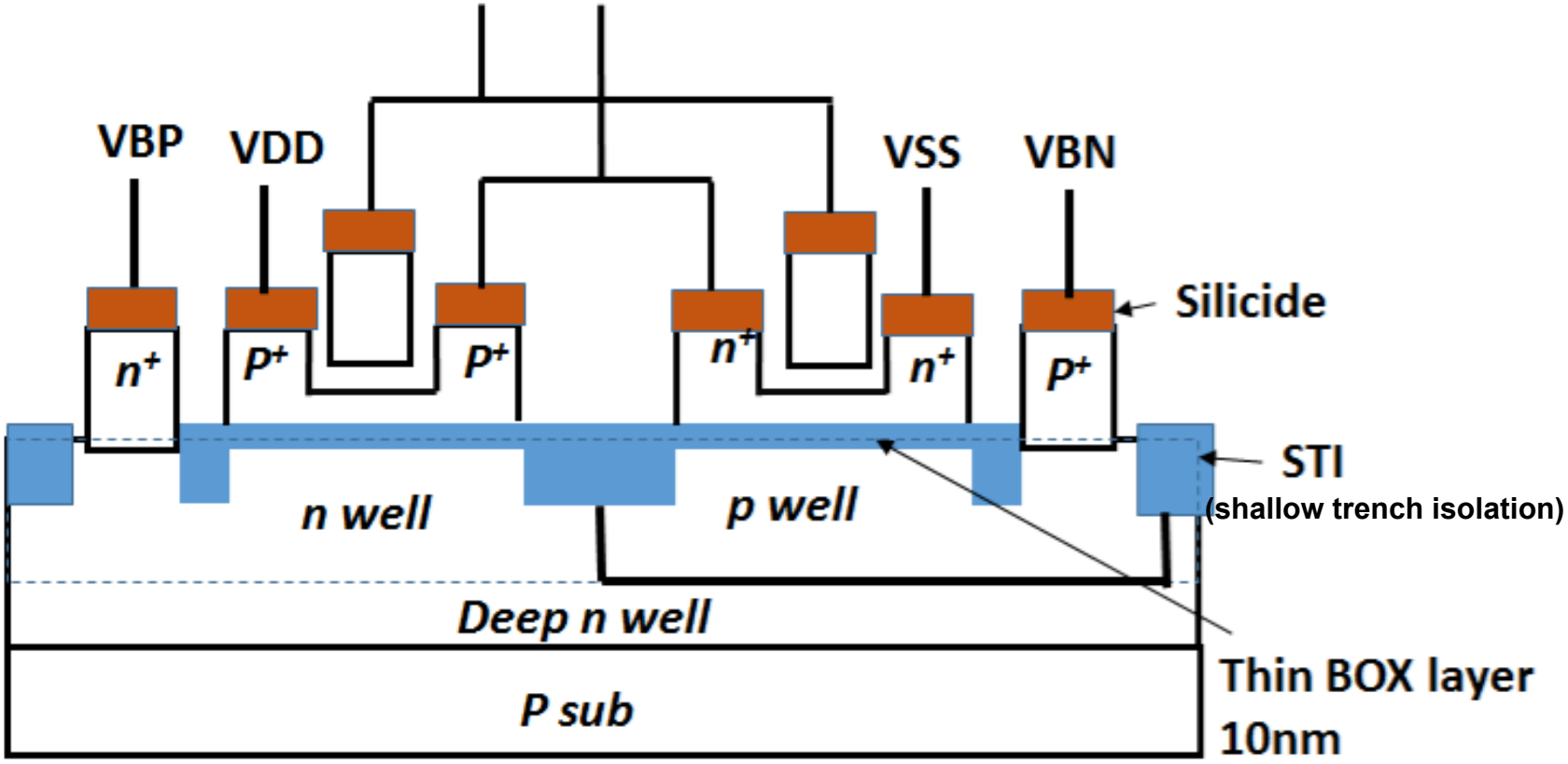
We used the SOTB technology to develop a coarse-grained reconfigurable accelerator.



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# SOTB CMOSFET




Cross-sectional view of the SOTB Device

# Characteristics of SOTB

1) Leakage power and delay can be widely controlled by body bias

2) SOI becomes more pronounced the lower the voltage operation and making high-speed operation possible

3) Parasitic thyristors are not formed in SOI

4) Anti-radiation tolerance is high

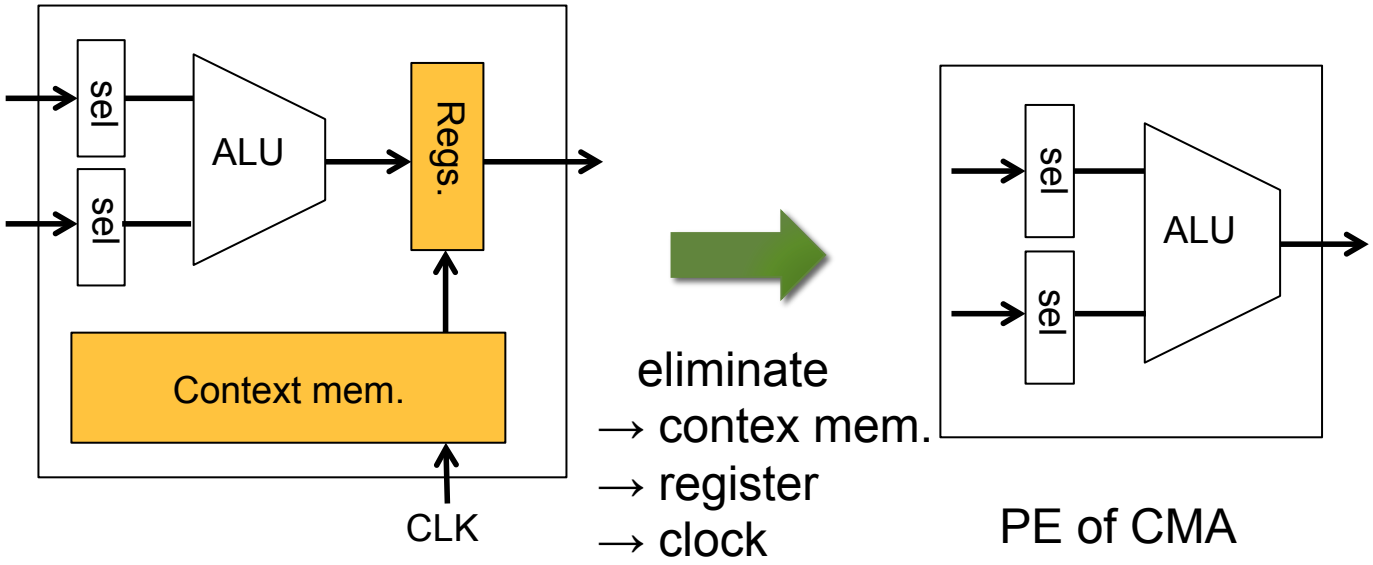
5) Noise propagation (cross-talk) is small

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# Concept of CMA

- Cool Mega Array(CMA) is one of coarse grained reconfigurable accelerators[1]
- Key concept is reducing any energy usage other than that required for computation.



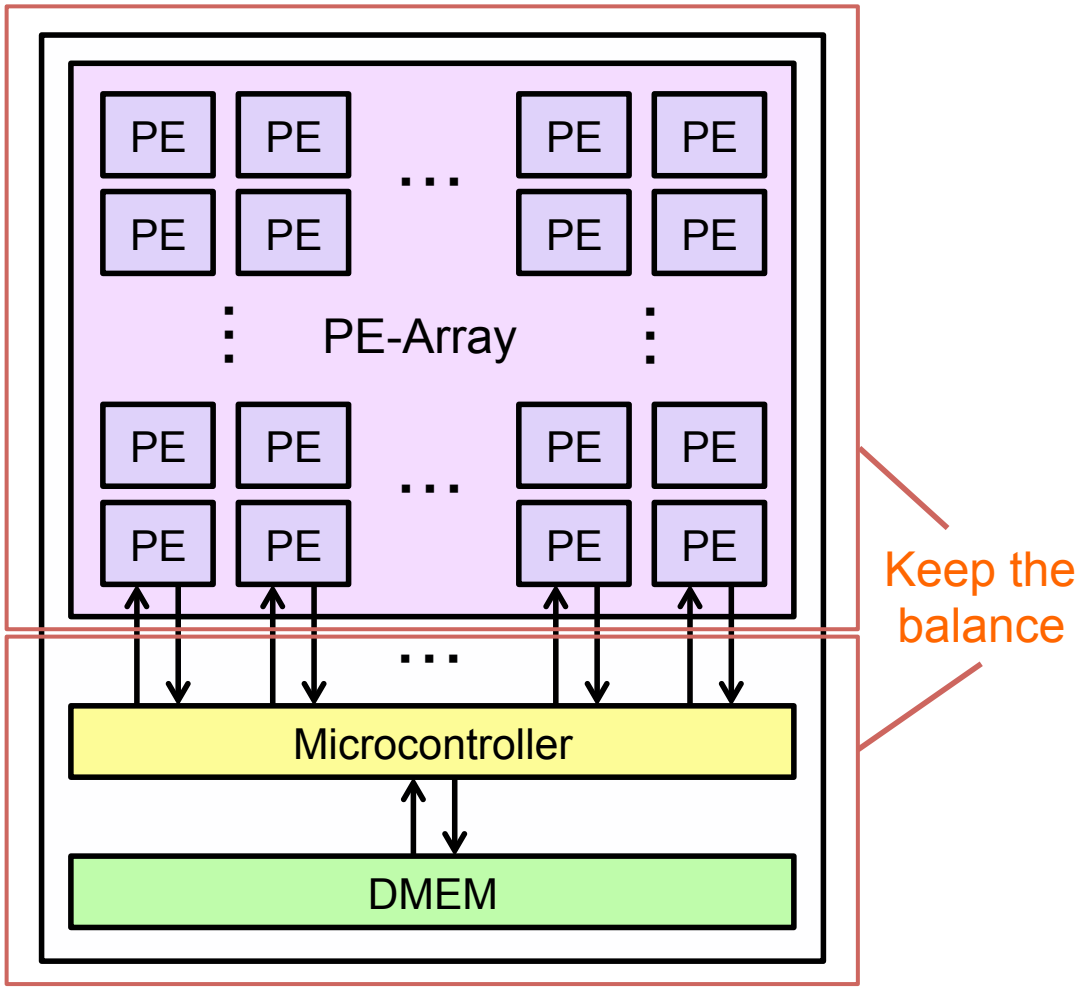

PE of CGDRP
(Coarse-Grained Dynamic Reconfigurable Processors)

PE of CMA

[1] N.Ozaki, et. al. “Cool Mega-Arrays : Ultralow-Power Reconfigurable Accelerator Chips” (IEEE Micro ’13)

# Block-diagram of CMA

- CMA consists of
  - PE-Array (8x8 PEs, here)
  - Microcontroller
  - DMEM
- Balance between PE-Array operation speed and Microcontroller memory management speed is important.
- CMA uses voltage scale control to keep this balance.



# Related work

Coarse -grain

- In our previous trial[2], balancing the performance of the computational module and memory access by body bias control

Fine -grain

- An FPGA is divided into several block, and the DVS is applied to each block[3]
- FLEX Power FPGA controls the back-gate bias by the bits of configuration data in order to optimize the delay and leakage power[4]

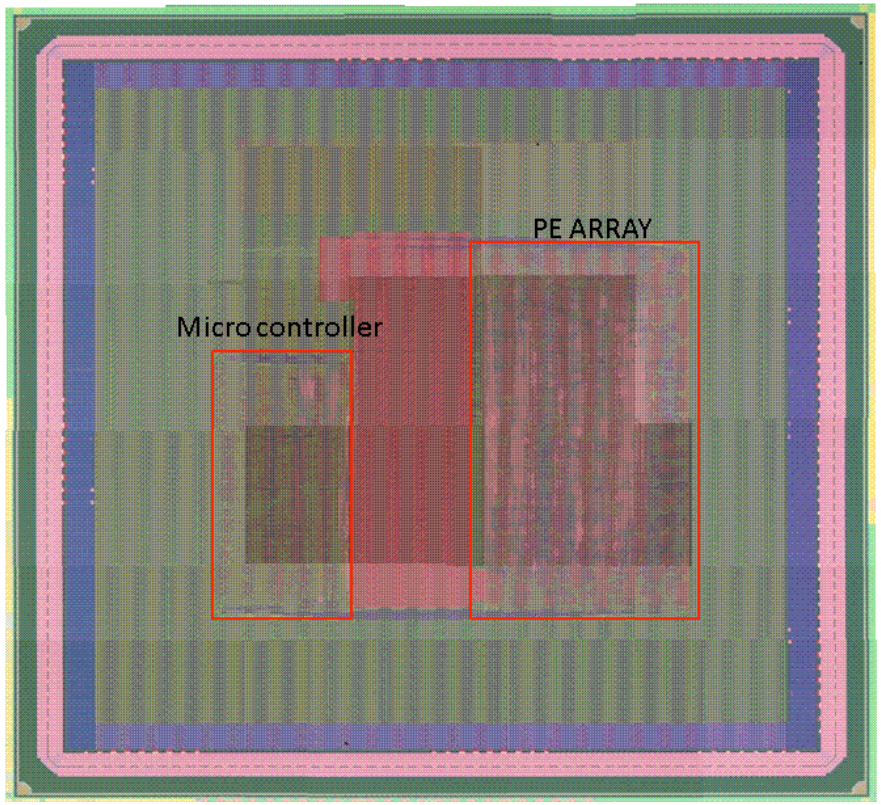
[2] Hongliang Su, et. al. “A Low Power Reconfigurable Accelerator using a Back-gate Bias Control Technique” (ICFPT ‘13)
[3] C.Q.Tran, et. al. “Low-Power Low-Leakage FPGA Design using Zigzag power-gating, dual-VTH/VDD and micro-VDD-hopping” (IEICE trans ‘06)
[4] M.Hioki, et. al “Fully-Functional FPGA Prototype with Fine-Grain Programmable Body Biasing” (ACM/SIGDA FPGA ‘13)

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# Design of CMA-SOTB

- For memory cells, registers are used.
- PE-Array and Microcontroller share the same VDD, but uses independent body bias




Chip photo of CMA-SOTB

| Specification of CMA-SOTB | | |
|---|---|---|
| Chip | Process | LEAP 65nm SOTB 7-metal |
| | Size | 5mm x 5mm |
| | I/O | 208pins |
| Tools | Design | Verilog HDL |
| | Synthesis | Synopsys Design compiler 2011.09-SP2 |
| | P&R | Synopsys IC Compiler 2010.12.SP5 |

# Body bias of CMA-SOTB

- Body bias
  - VBN is Bias voltage of N-MOS transistors
  - VBP is Bias voltage of  P-MOS transistors

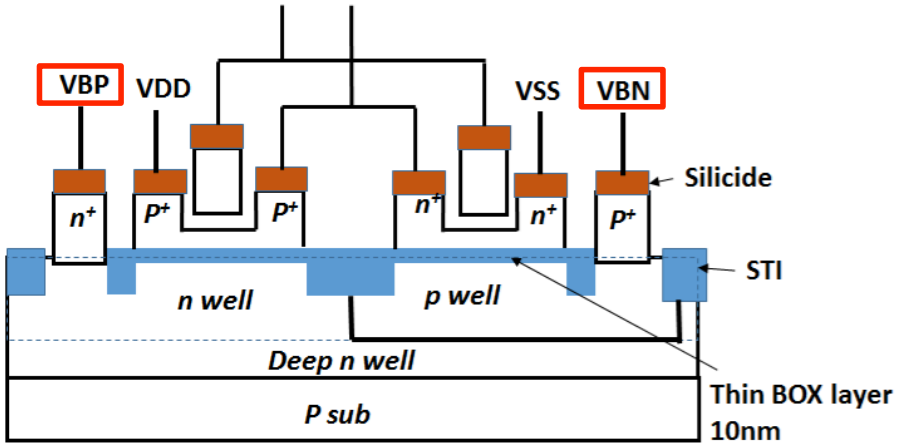
- VBN > 0, VBP < VDD

  → Forward bias

  Delay↓ Frequency↑ Leakage↑
- VBN < 0, VBP > VDD

  → Reverse bias

  Delay↑ Frequency↓ Leakage↓




Cross-sectional view of the SOTB Device

Performance ⟺ Leakage power

Trade-off

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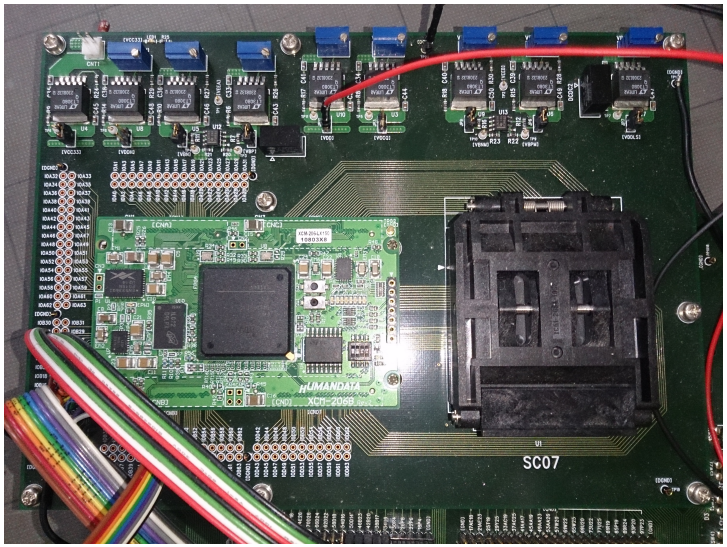
# Real chip evaluation

Investigate policy of body bias control

- PE-Array VBNC・VBPC
- Microcontroller VBNM・VBPM

- We gave same bias voltage to P-MOS(VBP) and N-MOS(VBN)
  → VBPC = VDD – VBNC
  → VBPM = VDD – VBNM

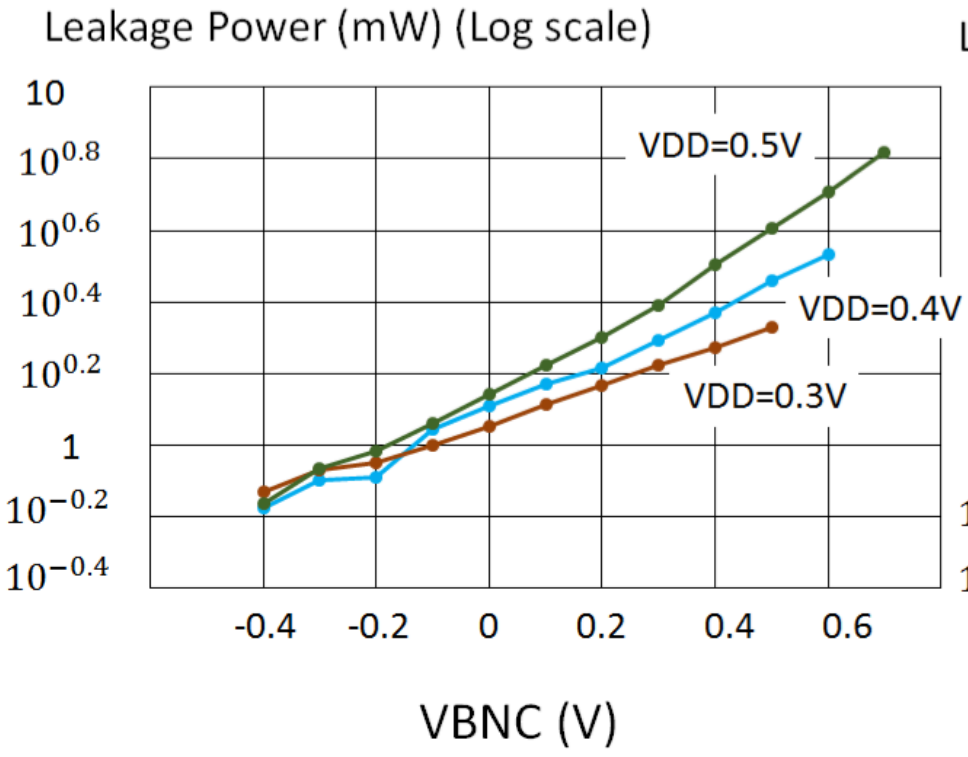
The bias is only represented with VBNC and VBNM.

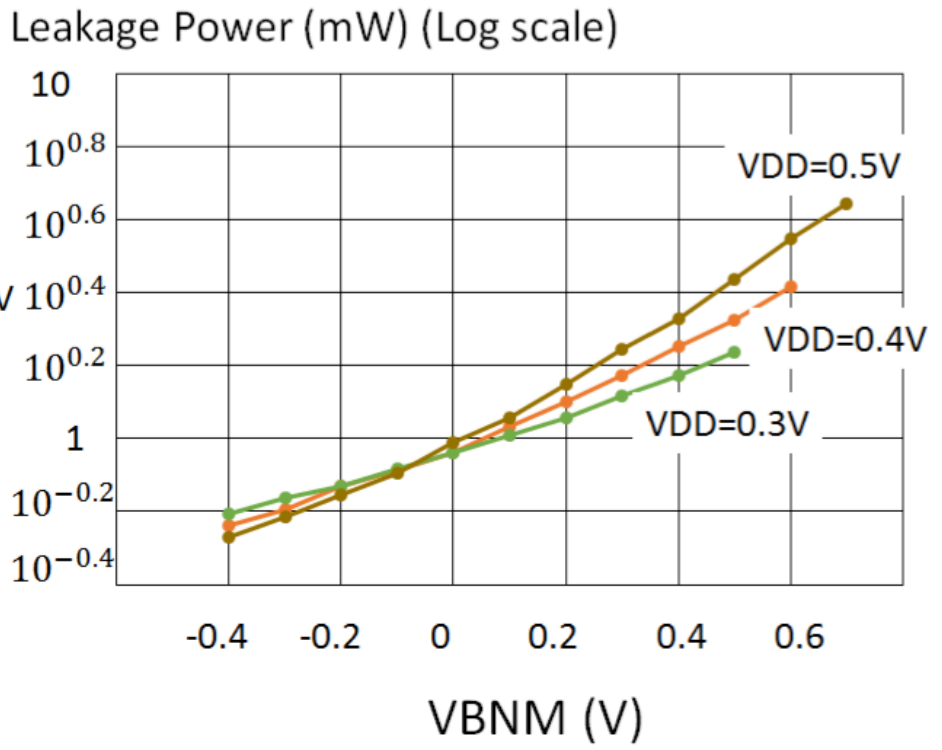


# Leakage power vs. Body bias voltage

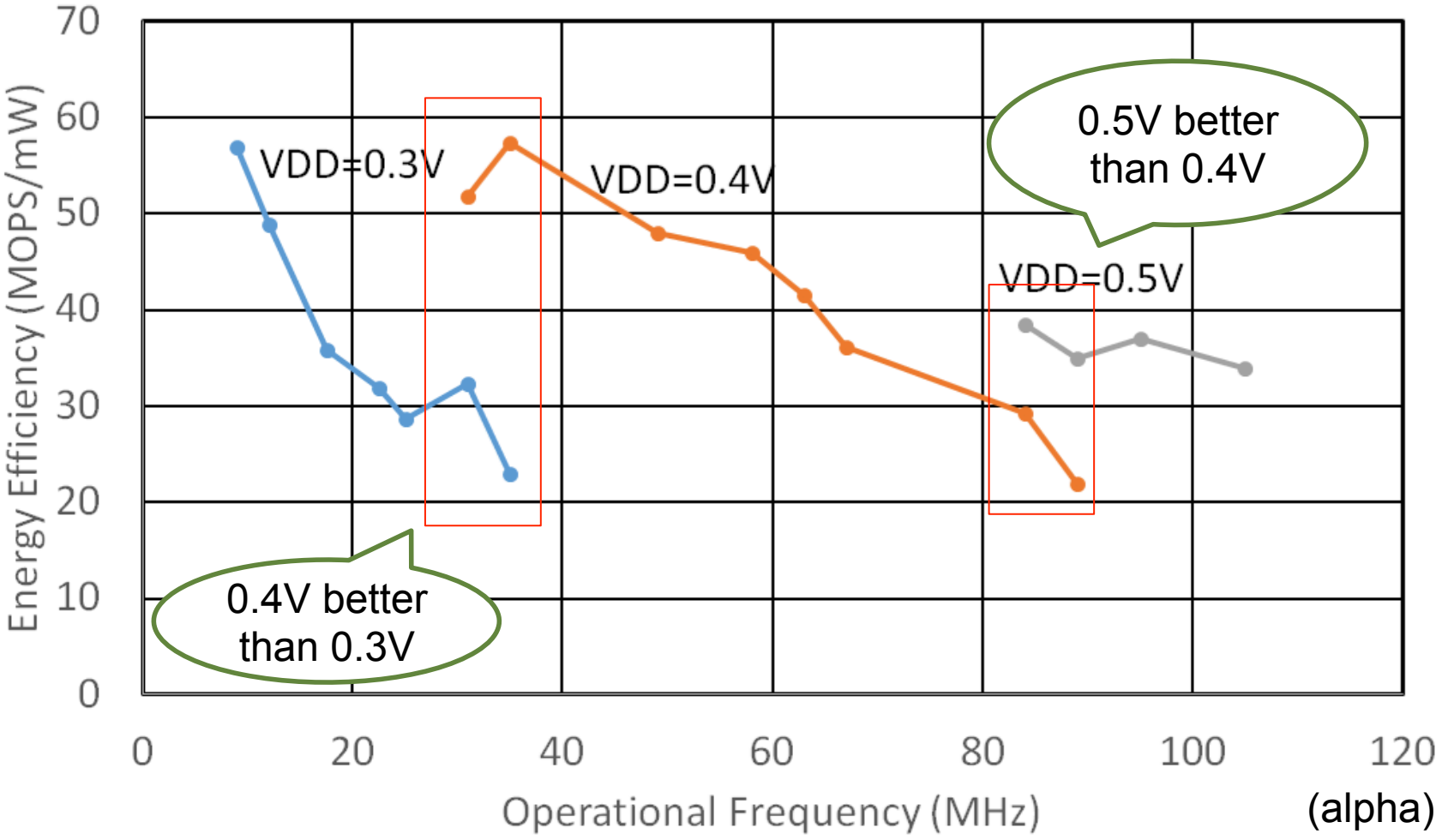
- Leakage power increases exponentially
- Leakage power of PE-Array is larger than that of the microcontroller

- VBNC = VBNM = -1.3V, VDD = 0.4V
  → Stand-by power 14.4μW



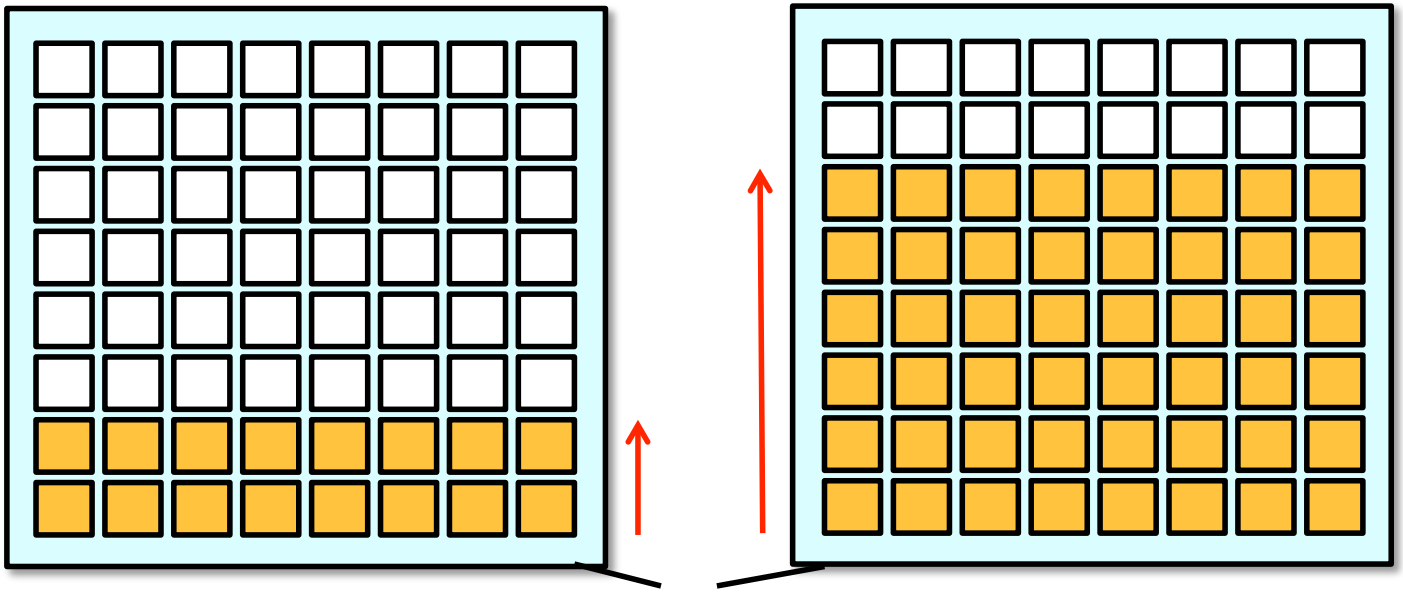
# Energy efficiency vs. operational frequency




- Supply Voltage is determined in accordance with the required performance

# Used program for the evaluation

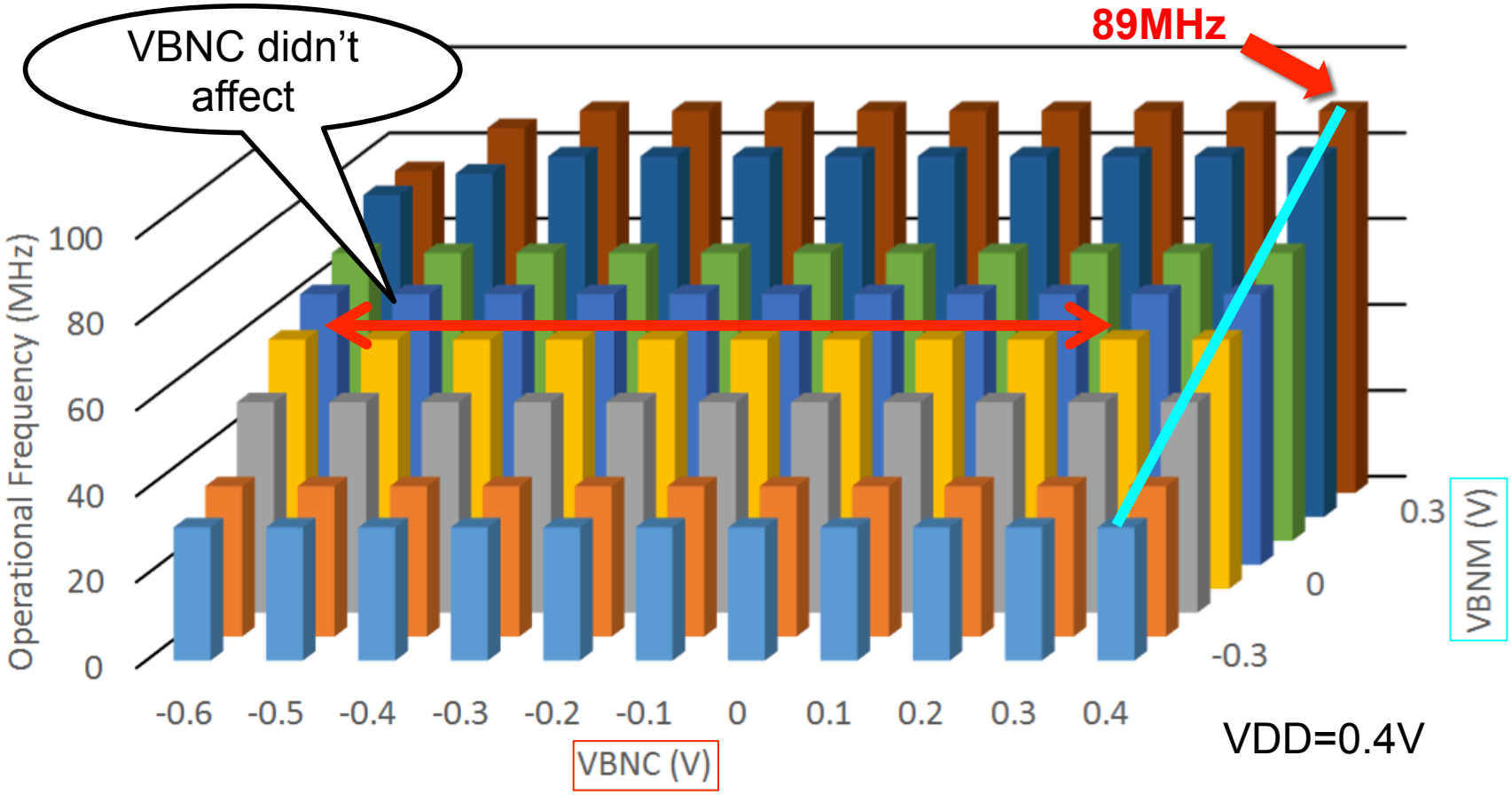
- Simple image processing programs alpha and packed alpha are used.
- Alpha blender of two images.




- Light-weight program
- Main operation is memory management

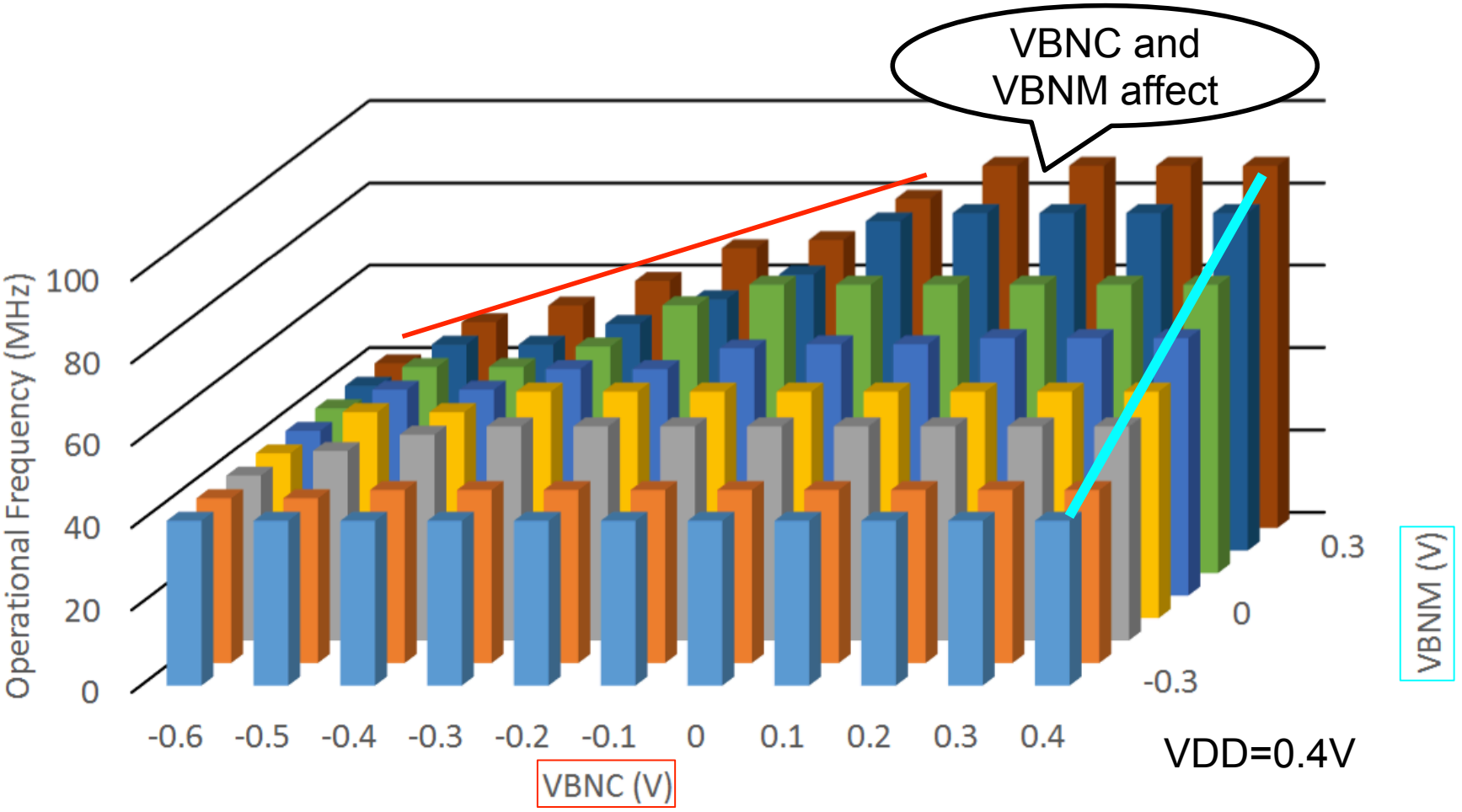
- Heavy-weight program
- Arithmetic intensity is high

# Operational frequency vs. Body bias voltage (alpha)



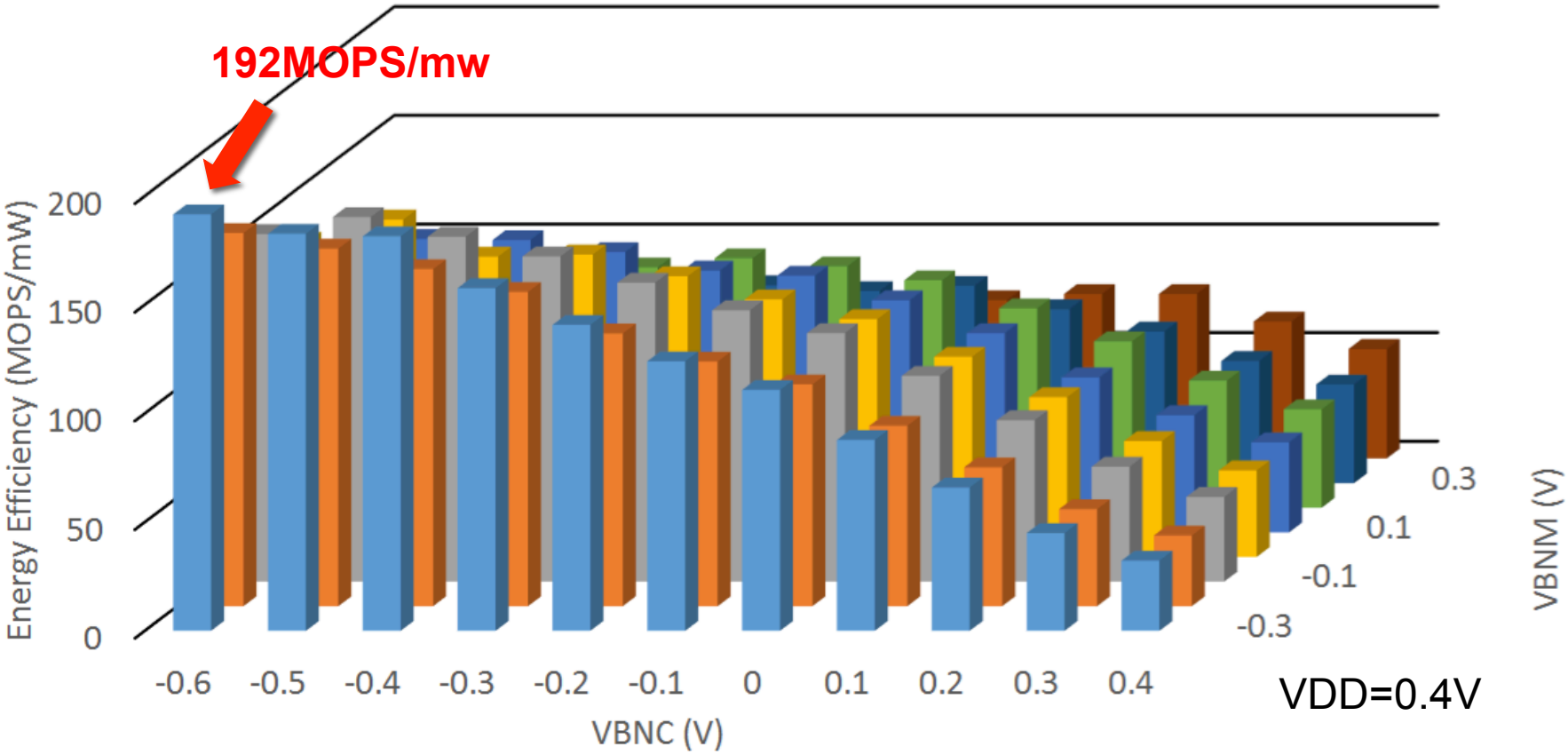

- Since the dataflow graph of alpha is light-weight, the operational frequency is mostly controlled by VBNM

# Operational frequency vs. Body bias voltage (packed alpha)




- In this application, both VBNC and VBNM influence the operational frequency.

# Energy efficiency vs. Body bias voltage(packed alpha)




- Energy efficiency is better when it works with lowest frequency clock
- 192MOPS/mW sustained energy efficiency was achieved.

※ MOPS = Million Operation Per Second

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# Policy of Body bias control

1) The optimal supply voltage range is fixed according to the required performance

→ Operational Freq. is

| | |
|---|---|
| over 80MHz | 0.5V |
| 80MHz~30MHz | 0.4V |
| under 30MHz | 0.3V |

2) Choose the combination of the smallest VBNC and VBNM that satisfy the required performance

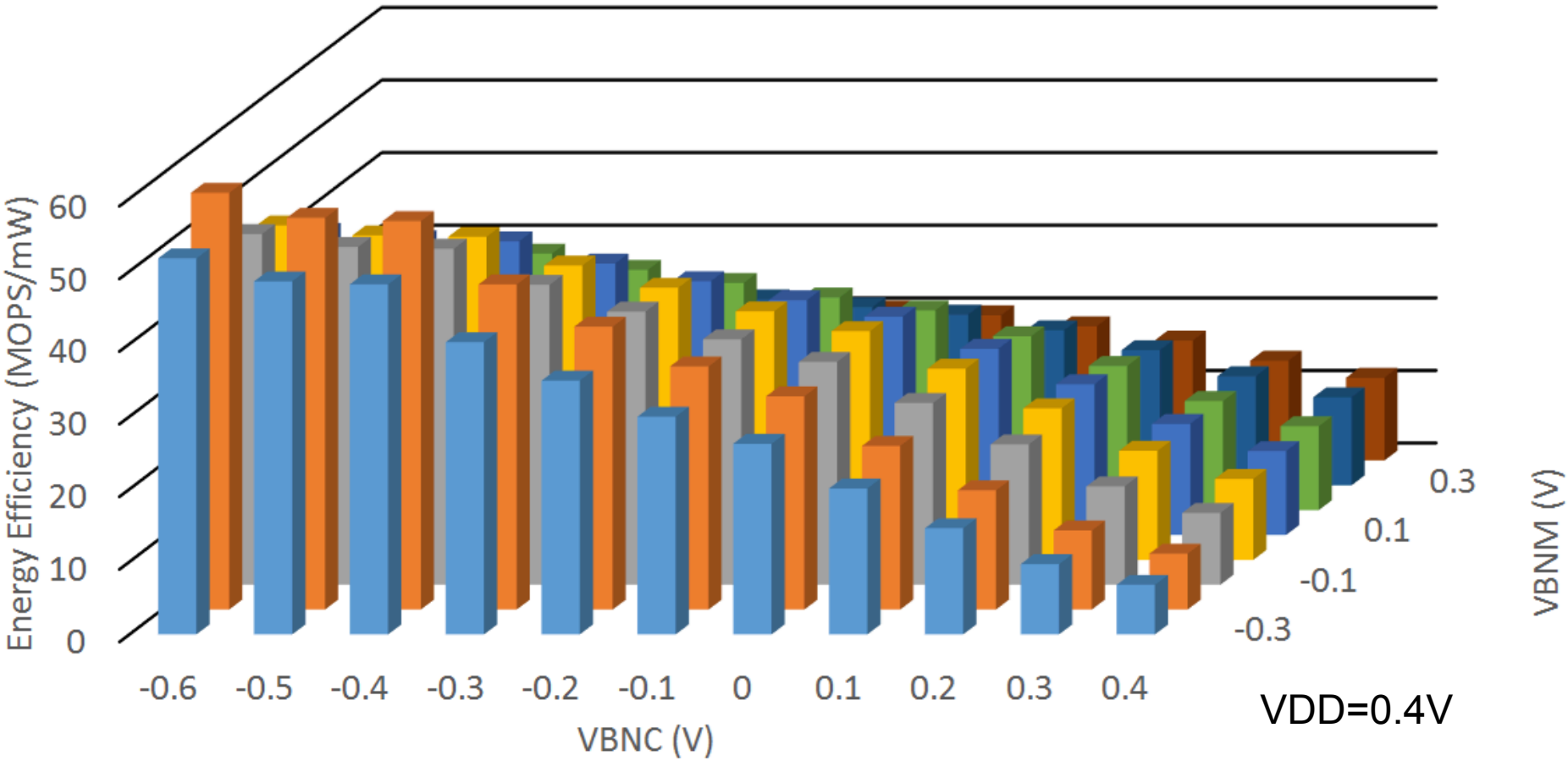
→ Leakage power increases exponentially with the body bias voltage

→ Set VBNC small as possible.

# Conclusion

- We developed and evaluated the CMA-SOTB
  - 192MOPS/mW energy efficiency is achieved with 0.4V supply voltage.
  - The maximum operational freq. of 89MHz was achieved with only 0.4V supply voltage.
  - Leakage power is smaller than 20μW in  the stand-by mode
- A policy to control the body bias was shown.

# Energy efficiency vs. Body bias voltage(alpha)




- Energy efficiency is more higher when lower the frequency by strong reverse bias

※ MOPS = Million Operation Per Second